

Eric Seo

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Education

Cornell University, College of Engineering, Ithaca, NY
Bachelors of Science, Electrical and Computer Engineering

Expected May 2028

GPA: 3.69/4.00

Relevant Experiences

CUXR (Cornell University eXtended Reality)

January 2025 – Present

Full Team Lead

Ithaca, NY

- Managing team of ~30 members creating multiple projects such as horror games in VR, custom edge detection for use in Meta Glasses, and haptic feedback systems mapping computer feedback to physical sensations.
- Contributed hands-on engineering work, including programming MCUs with C/C++, integration of custom hardware with Oculus Quest 2s, and the creation of a system releasing scents integrated with the VR world.
- Hosted information sessions with Valve and General Motors and working to secure sponsorships from various BCI and EEG companies.

Picogrid Hardware Engineering

May 2026 – August 2026

Hardware Intern

El Segundo, CA

- Owned end-to-end mixed-signal PCB design for complex 8-layer boards, including controlled-impedance differential routing, signal-integrity analysis, and EMI mitigation.
- Drove development and bring-up of USB-C Power Delivery and STM32-based hardware through rapid prototyping and precision microsoldering.
- Led project scoping from requirements through prototype, defining technical approaches and execution plans.

Kiewit Power Engineering

May 2025 – August 2025

Power Engineer

Olathe, KS

- Supported multiple projects at different stages including commissioning and plant closeout for electrical systems.
- Performed circuit design, cable sizing, and voltage drop calculations to ensure systems met electrical standards.
- Updated and maintained key documentation including one-line diagrams and schematics to keep records accurate and aligned with field conditions.

Projects

Omnidirectional Treadmill

April 2026 - Present

Cornell XR

- Led 360° platform design, setting mechanical, structural, and user-safety requirements for VR treadmill use.
- Designing a buck-converter PCB and MCU control system to drive treadmill motion from live camera input.
- Developing a CMU OpenPose pipeline for real-time user motion tracking and closed-loop treadmill control.

DEX (Dextrous Hand)

February 2026

Cornell Makeathon 2026 (Best Hardware Winner)

- Designed DEX, a wearable hand-tracking peripheral for real-time motion sensing and tactile interaction.
- Built the full embedded electronics stack, wiring up dual rotary encoders, a 9-DOF IMU, and a thumb switch to an ESP32 and getting motion data over I²C/SPI.
- Wrote firmware to read calibrated sensor data to a web platform as usable finger-tracking data for rehab and accessibility applications.

Skills

Relevant Coursework: Circuits and Circuit Analysis, Digital Logic & Computer Organization, Microelectronics, Signals and Systems, EM Fields and Waves, Embedded Systems, Linear Algebra, Differential Equations

Skills/Tools: KiCad, EMI Analysis, PCB Design, C++, C, Python, SystemVerilog, LTSpice, LaTeX, Git/Github, STM32